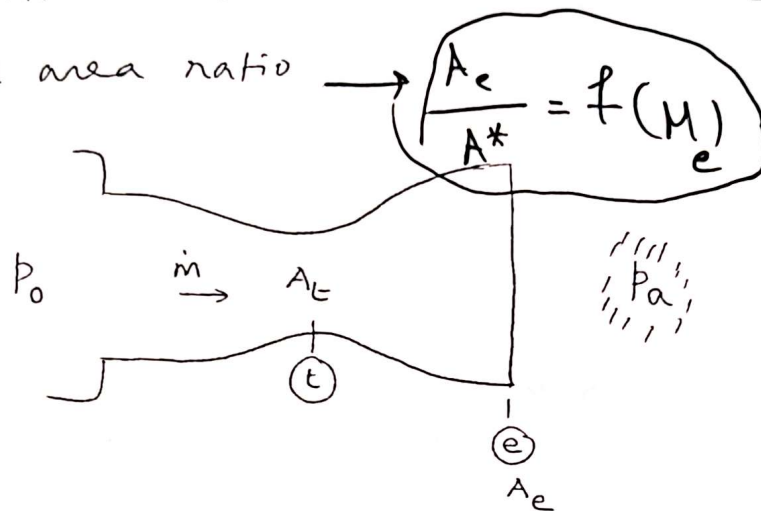


Flow behavior in c-d nozzles

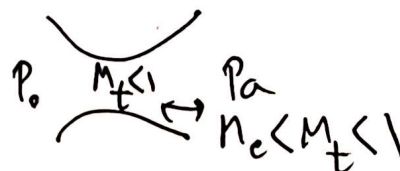
The Mach number at the exit of a c-d nozzle is governed by two factors

- ① the difference between the pressure in the regions just upstream and downstream of the nozzle; more precisely, the ratio of the pressures $\rightarrow \frac{p_e}{p_o}$
- ② The nozzle area ratio $\rightarrow \frac{A_e}{A^*} = f(M_e)$

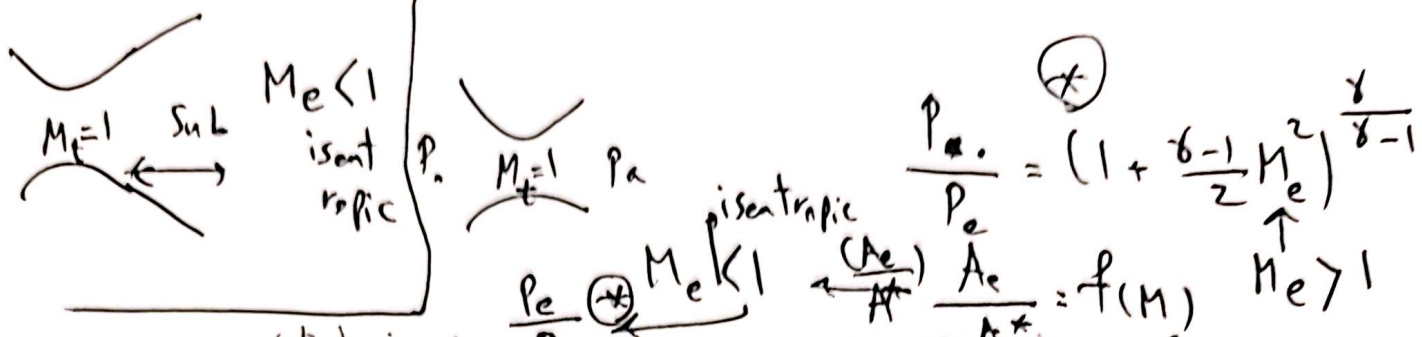


In the figure, the c-d nozzle is between regions with pressures p_o and p_a , and has area ratio $\frac{A_e}{A_t}$. Thus, the values of (p_o/p_a) and (A_e/A_t) govern the Mach number at the nozzle exit (M_e).

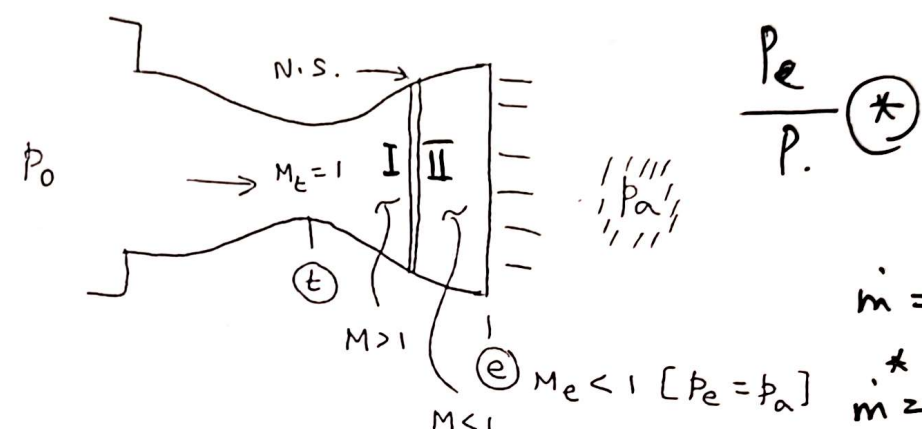
* If $\left(\frac{p_o}{p_a}\right)$ is ^{$\frac{p_a}{p_o}$ high} low, then the flow through the nozzle will be entirely subsonic, and both M_t and M_e will be less than 1. The highest Mach number will occur at the throat. Since $M_e < 1$, $p_e = p_a$.



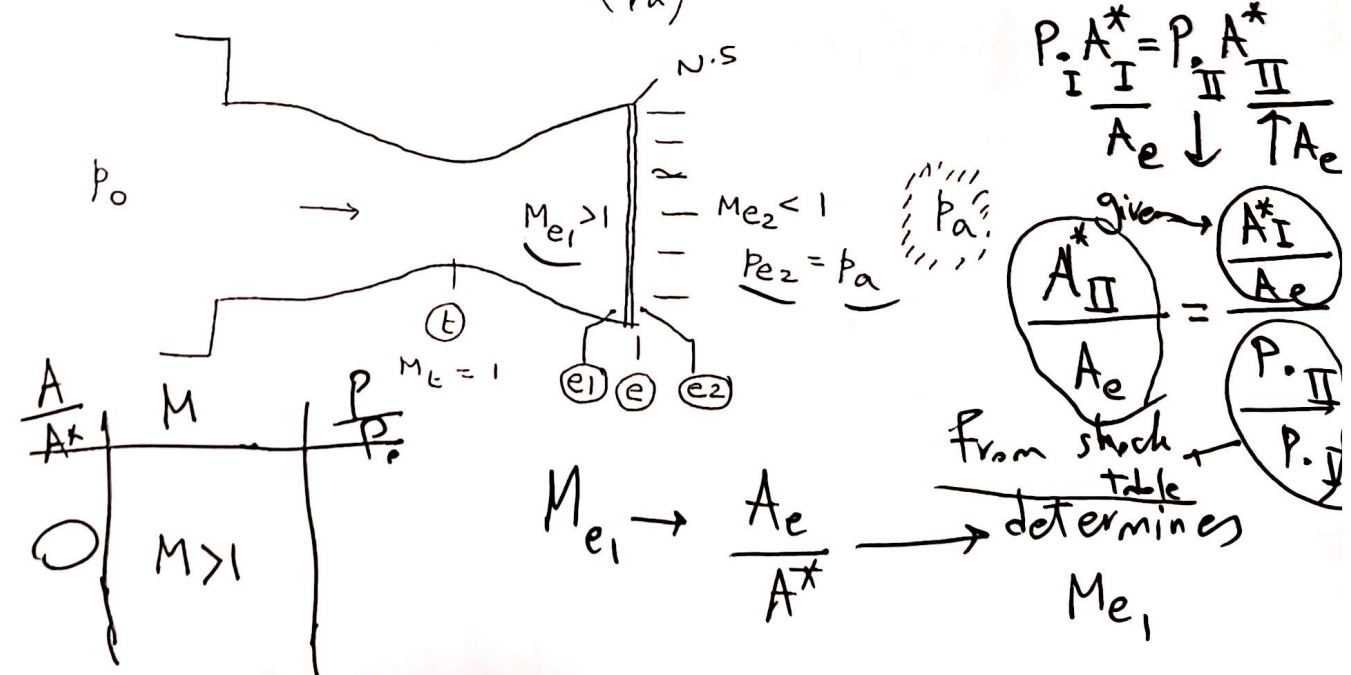
p_a is not much smaller than p_o



* If $\left(\frac{p_0}{p_a}\right)$ is large, $M_e = 1$, and the flow will accelerate to supersonic velocities in the diverging portion of the nozzle before encountering a normal shock. The flow then becomes subsonic and $M_e < 1$.



* If $\left(\frac{p_0}{p_a}\right)$ further increases, the normal shock will move downstream and will station itself at the exit plane at a specific $\left(\frac{p_0}{p_a}\right)$ value.



The value of $\left(\frac{p_0}{p_a}\right)$ at this condition is, say, $\left(\frac{p_0}{p_a}\right)_{crit}$,
and will be a function of nozzle area ratio $\left(\frac{A_e}{A_t}\right)$.

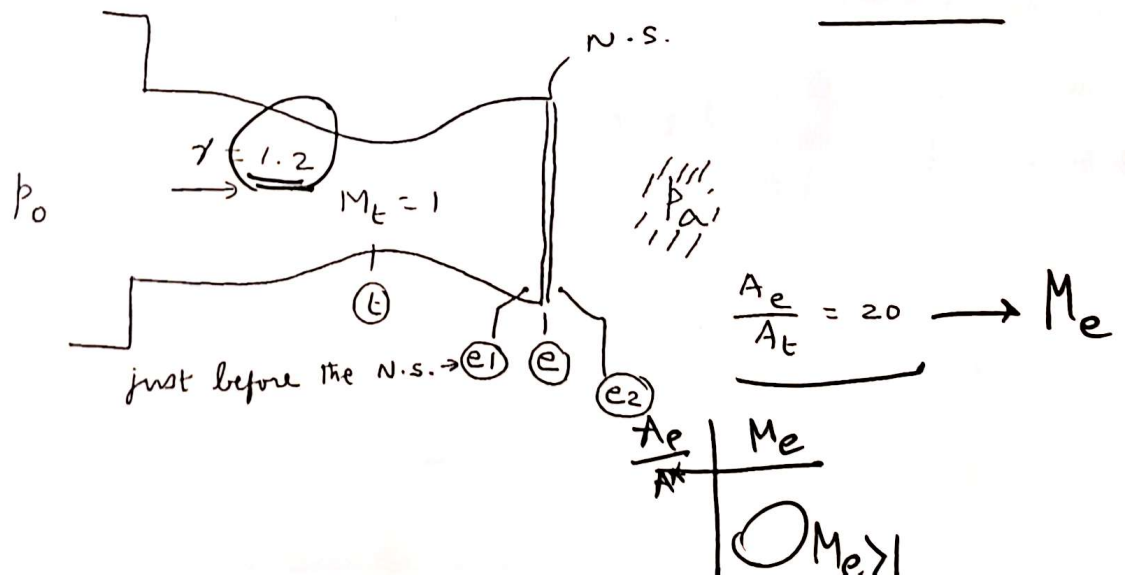
* If $\left(\frac{p_0}{p_a}\right)$ is increased further, the shock will move outside the nozzle, and $M_e > 1$. The flow will continuously accelerate from $M < 1$ at the upstream reservoir, reach $M_t = 1$ at the throat, and leave the nozzle at $M_e > 1$.

* Thus the minimum $\left(\frac{p_0}{p_a}\right)$ required to have a shock-free flow in the c-d nozzle is $\left(\frac{p_0}{p_a}\right)_{crit}$.

Example 1.2

A c-d nozzle has $\frac{A_e}{A_t} = 20$. Find $\left(\frac{p_0}{p_a}\right)_{crit}$.

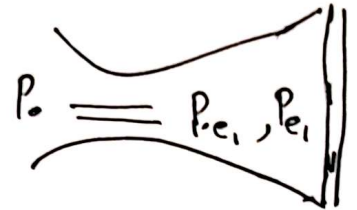
Assume that the nozzle is operating with $\left(\frac{p_0}{p_a}\right) = \left(\frac{p_0}{p_a}\right)_{crit}$



$$A_{e1} = A_e \Rightarrow \frac{A_{e1}}{A_t} = 20$$

$$\frac{A_{e1}}{A^*} \xrightarrow{M_{e1}=1} \frac{A_{e1}}{A_t} = \underline{\underline{20}}$$

$$\frac{A_{e1}}{A^*} = \frac{1}{M_{e1}} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_{e1}^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$



Solve for M_{e1} by trial-and-error, using MATLAB or online gas dynamic calculator. Observe $M_{e1} > 1$.

$$M_{e1} = 3.76 \quad (\gamma = 1.2) \quad \text{Isen, no work}$$

$$p_{e1} = \frac{p_{0e1}}{\left(1 + \frac{\gamma-1}{2} M_{e1}^2 \right)^{\frac{\gamma}{\gamma-1}}} = \frac{p_{0e1}}{197.78} = \frac{p_0}{197.78}$$

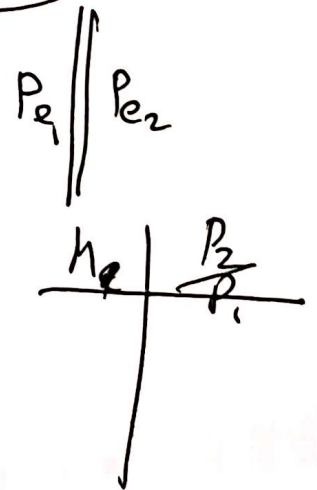
Use the normal shock relation,

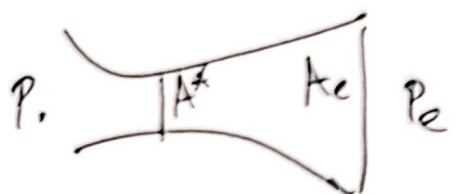
$$\frac{p_{e2}}{p_{e1}} = \frac{2\gamma}{\gamma+1} M_{e1}^2 - \frac{\gamma-1}{\gamma+1}, \quad \text{N.S.}$$

online normal shock calculator to get

$$\frac{p_{e2}}{p_{e1}} = \underline{\underline{15.33}} \quad (M_{e1} = 3.76, \gamma = 1.2)$$

$$p_{e2} = \underline{\underline{15.33}} p_{e1} = \underline{\underline{15.33}} \cdot \frac{p_0}{197.78} = \underline{\underline{0.0775 p_0}}$$





$$M_{e2} < 1 \Rightarrow p_{e2} = p_a$$

$$\frac{p_0}{p_e} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}} = 197.78$$

$$p_a = 0.0775 p_0 \Rightarrow \frac{p_0}{p_a} = \frac{1}{0.0775} = 12.90$$

This is the $\left(\frac{p_0}{p_a}\right)_{crit}$ value: $\left(\frac{p_0}{p_a}\right)_{crit} = 12.90$

Observe that this result depends on the nozzle area ratio; 20 in this case. If $\left(\frac{A_e}{A_t}\right)$ had been different, $\left(\frac{p_0}{p_a}\right)_{crit}$ would have been different as well.

1.2 → Example: An ideal rocket thrust chamber has $p_0 = 1 \text{ MPa}$ and nozzle area ratio of $\frac{20}{1}$. Will the nozzle be shock-free during operation at 10-km altitude? What are the exit Mach number and the exit pressure? $\gamma = 1.2$.

At 10-km altitude, $p_a = 26.5 \text{ kPa}$

$p_0 = 1 \text{ MPa} = 1000 \text{ kPa}$

$\left(\frac{p_0}{p_a}\right)_{actual} = \frac{1000}{26.5} = 37.74$

The nozzle area ratio = 20, for which $\left(\frac{p_0}{p_a}\right)_{crit} = 12.90$

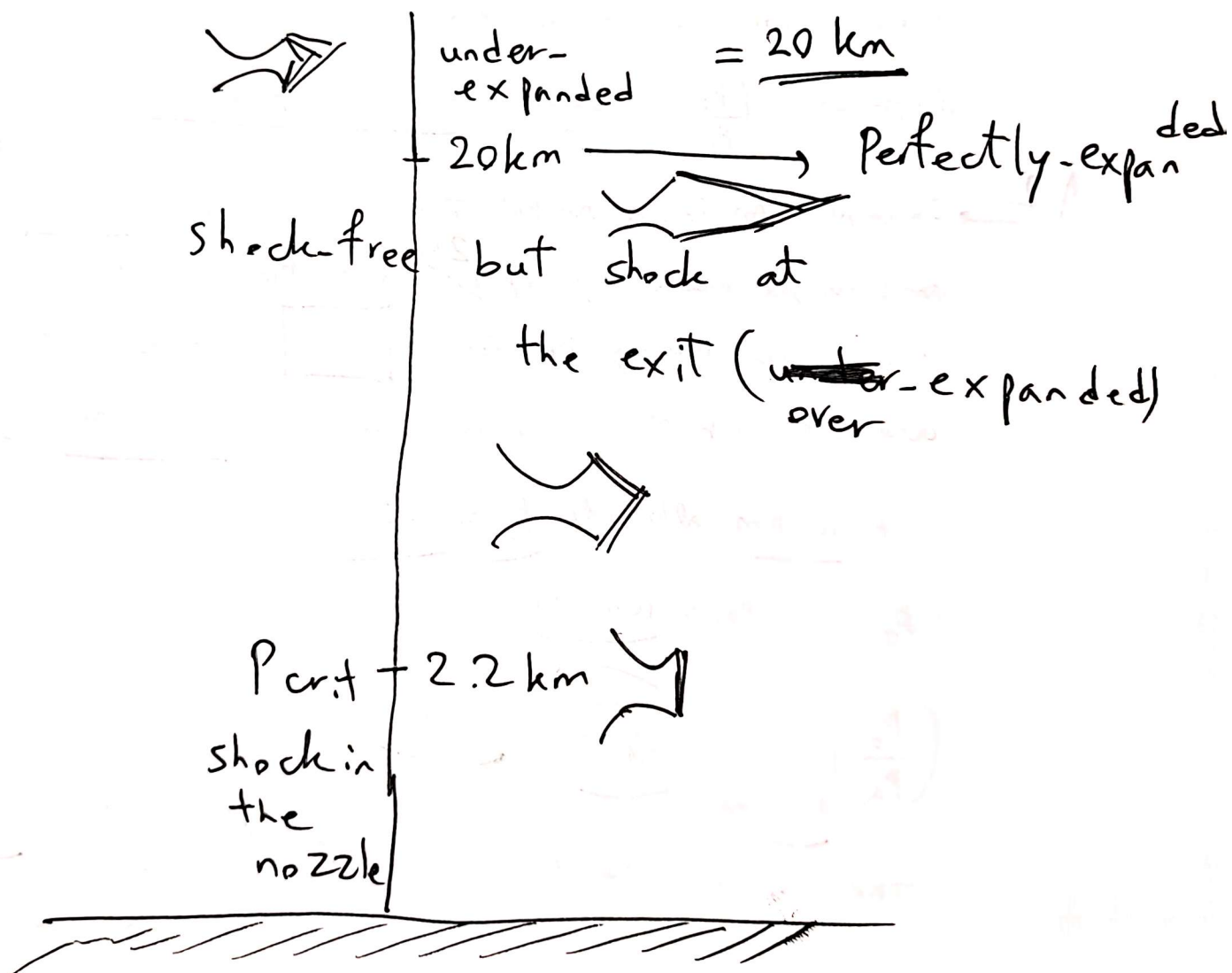
$\left(\frac{p_0}{p_a}\right)_{actual} > \left(\frac{p_0}{p_a}\right)_{crit} \Rightarrow$ shock free nozzle flow.

$$\frac{P_0}{P_e} = 197.78 \quad \text{perfectly expanded}$$

Perfectly-expanded

$$P_0 = 1 \text{ MPa} \longrightarrow P_e = \frac{1 \text{ MPa}}{197.78} = 5.06 \text{ kPa}$$

$$h (p = 5.06 \text{ kPa})$$



The exit Mach number is fixed by area ratio (20).

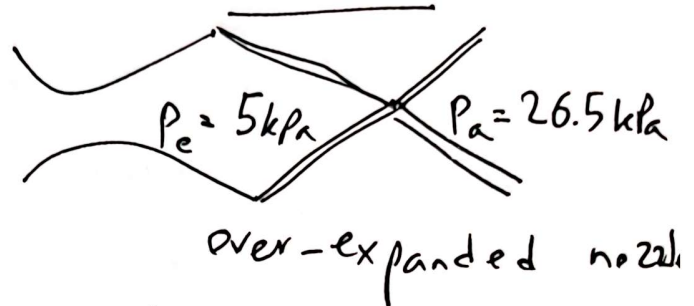
This was already found. $M_e = 3.76$

$$p_e = \frac{p_{0e}}{\left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}} = 5056 \text{ Pa} = 5.056 \text{ kPa}$$

isentropic $p_0 = 1 \text{ MPa}$

$$p_a = 26.5 \text{ kPa}$$

$p_e < p_a$ (overexpanded jet)



Example: What is the lowest altitude where the thrust chamber of the previous example operates shock free?

$$\left(\frac{p_0}{p_a}\right)_{\text{crit}} = 12.90$$

$$p_0 = 1 \text{ MPa} \Rightarrow \left(p_a\right)_{\text{crit}} = \frac{10^6}{12.90} = 77,519 \text{ Pa}$$

From atmospheric table (Table 1, p 700) and interpolating,

$$h_{\text{crit}} = 2211 \text{ m} \approx 2.2 \text{ km}$$

The lowest altitude for shock-free operation is 2.2 km.

At $h < h_{\text{crit}}$, there will be a shock in the

diverging portion of the c-d nozzle.

Comment

Observe that, for nozzle flow with $\left(\frac{p_0}{p_a}\right)_{\text{actual}}$ just greater than

$\left(\frac{p_0}{p_a}\right)_{\text{crit}}$, $\underline{p_e} < \underline{p_a}$. To attain the fully expanded

condition ($\underline{p_e} = \underline{p_a}$), $\left(\frac{p_0}{p_a}\right)$ has to be further increased.

Thus, if the flow is stated to be fully expanded or

underexpanded ($\underline{p_e} > \underline{p_a}$), it follows that $\left(\frac{p_0}{p_a}\right)_{\text{actual}} > \left(\frac{p_0}{p_a}\right)_{\text{crit}}$,

and there can be no shocks in the nozzle.

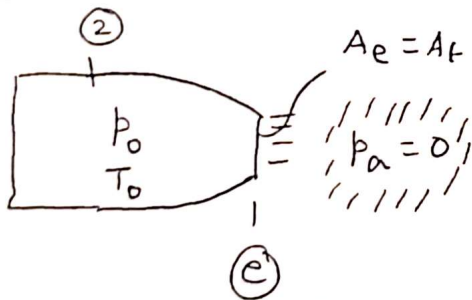
$$P_0 = 1.0 \text{ MPa}$$

Example: Plm 11.2

An ideal rocket thrust chamber has $P_0 = 10 \text{ MPa}$ and $T_0 = 3000 \text{ K}$. $\gamma = 1.2$, $\bar{M} = 25$, $A_t = 0.1 \text{ m}^2$. Calculate M_e , u_e , p_e and $\dot{m}$, and c^* and C_f for each of the following cases:

- ① converging nozzle, $P_a = 0$
 - ② c-d nozzle ($A_e = 4.06 \text{ m}^2$), $P_a = 0$
 - ③ converging nozzle, $P_a = 0.1 \text{ MPa}$
 - ④ c-d nozzle ($A_e = 4.06 \text{ m}^2$), $P_a = 0.1 \text{ MPa}$
- $$R = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{25} = 332.6 \frac{\text{J}}{\text{kg} \cdot \text{K}}, \quad \gamma = 1.2$$

Case ①



conv. nozzle: $\frac{A_e}{A_t} = 1$

Let $P_e = P_a = 0$
 $P_{0e} = P_0 = 10 \text{ MPa}$
 isen $\Rightarrow \frac{P_{0e}}{P_e} \rightarrow \infty \neq P_a$

$\left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}} \rightarrow \infty \Rightarrow M_e \rightarrow \infty$

$M_e \neq 1 \Rightarrow M_e = 1$
 conv. nozzle.

$P_e = \frac{P_{0e}}{\left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}} = 5.645 \text{ MPa}$
 $M_e = 1$



$$T_e = \frac{T_{0e}}{1 + \frac{\gamma-1}{2} M_e^2} = 2727 \text{ K} \quad \left\{ T_{0e} = T_0 = 3000 \text{ K} \right\}_{\text{isen}}$$

$$u_e = M_e \sqrt{\gamma R T_e} = 1043 \text{ m/s}$$

$$\rho_e = \frac{p_e}{R T_e} = \frac{5.645 (10^6)}{(332.6)(2727)} = 6.224 \frac{\text{kg}}{\text{m}^3}$$

$$A_e = A_t = 0.1 \text{ m}^2$$

$$\dot{m} = \rho_e u_e A_e = 649.2 \text{ kg/s}$$

$$T_{\text{conv}} = \underset{649.2}{\dot{m}} \underset{1043 \text{ m/s}}{u_e} + (\underset{5.645}{p_e} - \underset{0}{p_a}) \underset{0.1}{A_e}$$

$$= (0.6771 + 0.5645) = 1.242 \text{ MN}$$

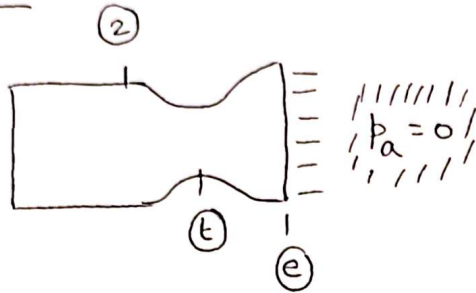
$$C^* = \frac{p_0 A_t}{\dot{m}} = \frac{(10)(10^6)(0.1)}{649.2} = 1540 \text{ m/s}$$

$$\left\{ \text{From Eqn (11.9)}, C^* = 1540 \text{ m/s} \right\}$$

$$C_{T_{\text{conv}}} = \frac{T_{\text{conv}}}{p_0 A_t} = \frac{1.242}{(10)(0.1)} = 1.242$$

$$\left\{ \text{From Eqn (11.12)}, C_{T_{\text{conv}}} = 1.24, p_a = 0 \right\}$$

Case (2)



$$A_e = 4.06 \text{ m}^2$$

$$\frac{A_e}{A_t} = \frac{4.06}{0.1} = 40.6$$

$$\frac{A_e}{A^*} = \frac{A_e}{A_t} = 40.6$$

$M_e = 1$

$$\frac{A_e}{A^*} = \frac{1}{M_e} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_e^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Trial-and-error, or online calculator $\Rightarrow M_e = 4.25$

$$p_e = \frac{p_{0e}^{\text{isen}}}{1 + \frac{\gamma-1}{2} M_e^2} = 0.020476 \text{ MPa}$$

$p_0 = 10 \text{ MPa}$

$$T_e = \frac{T_{0e}^{\text{isen}}}{1 + \frac{\gamma-1}{2} M_e^2} = 1069 \text{ K}$$

$T_0 = 3000 \text{ K}$

$$u_e = M_e \sqrt{\gamma R T_e} = 2776 \text{ m/s}$$

$$\rho_e = \frac{p_e}{R T_e} = \frac{0.020476 (10^6)}{(332.6)(1069)} = 0.05759 \frac{\text{kg}}{\text{m}^3}$$

$$\dot{m} = \rho_e u_e A_e = 649.1 \text{ kg/s}$$

$$c^* = \frac{p_0 A_t}{\dot{m}} = \frac{10 (10^6) (0.1)}{649.1} = 1540 \text{ m/s}$$

$$(11.9) \Rightarrow c^* = 1540 \text{ m/s}$$

$$\frac{P_0}{P_e} = 197.78 \quad \text{perfectly expanded}$$

Perfectly-expanded

$$P_0 = 1 \text{ MPa} \longrightarrow P_e = \frac{1 \text{ MPa}}{197.78} = 5.06 \text{ kPa}$$

$$h (p = 5.06 \text{ kPa})$$

